

Regatta Structure Import from Manage2Sail

This is the documentation of the import of regatta structures from Manage2Sail.com. It contains information about the content of the XRR document as well as how the code realizes the import.

Through the import several regattas are created and connected with an existing event. It is also possible to create a new event for the regattas. Furthermore leaderboard groups will be created from the application. There is the possibility to create competitors for all regattas in the system. But it is not yet possible to create them only for the selected regattas, you want to import.

Receiving Information from Manage2Sail

Manage2Sail provides a JSON document for each event. In this document the event name and all regattas of an specific event are listed. The regattas contain information about the boat class and link to an XML document, which contains the following information:

- Person
 - Gender
 - NOC
 - Given Name
 - Family Name
 - IFPersonID
 - PersonID
- Boat
 - Sail Number
 - Boat Name
 - BoatID
- Team
 - Gender
 - BoatID
 - Team Name
 - TeamID
 - Crew
 - PersonID
 - Position
- Event
 - Type
 - Title
 - IFEventID
 - EventID
 - Division
 - Gender
 - Title
 - IFClassicID
 - DivisionID
 - Race
 - RaceID
 - Race Number
 - Race Name

From the XRR Document we can get the following information being relevant for the regattas to be imported:

- Event name: JSON document -> event name
- Boat class: JSON document -> regatta -> boat class
- Regatta name: XRR document -> event -> title
- Races: XRR document event -> races

Because the races in the XRR document are just listed with the race name and the race number, you can't see directly if there are any series or if there are fleets in the regatta. Therefore we made up some heuristics to see, whether there are series or fleets.

Series:

In all XRR documents we saw, the race name consists at least of a number with a prefix. The prefix consists of one character:

```
RaceNumber="1" RaceName="Q1 Yellow"/>
} RaceNumber="1" RaceName="Q1 Blue"/>
" RaceNumber="2" RaceName="Q2 Blue"/>
} RaceNumber="4" RaceName="Q4 Blue"/>
} RaceNumber="4" RaceName="Q4 Yellow"/>
} RaceNumber="3" RaceName="Q3 Blue"/>
" RaceNumber="2" RaceName="Q2 Yellow"/>
" RaceNumber="3" RaceName="Q3 Yellow"/>
RaceNumber="5" RaceName="F1 Gold"/>
" RaceNumber="6" RaceName="F2 Gold"/>
RaceNumber="5" RaceName="F1 Silver"/>
" RaceNumber="7" RaceName="F3 Silver"/>
} RaceNumber="6" RaceName="F2 Silver"/>
" RaceNumber="8" RaceName="M1 Medal"/>
} RaceNumber="7" RaceName="F3 Gold"/>
} RaceNumber="9" RaceName="M2 Medal"/>
RaceNumber="10" RaceName="M3 Medal"/>
```

Our heuristic to determine the series is to analyze the prefix. Each character stands for one series:

- Q: Qualification
- F: Final
- M: Medal
- R: Opening Series

If there is only one series, the series in the system is set to "default":

```
} RaceNumber="1" RaceName="R1"/>
} RaceNumber="4" RaceName="R4"/>
} RaceNumber="2" RaceName="R2"/>
} RaceNumber="3" RaceName="R3"/>
} RaceNumber="5" RaceName="R5"/>
} RaceNumber="6" RaceName="R6"/>
} RaceNumber="7" RaceName="R7"/>
```

Fleets:

To see whether a regatta is structured into fleets, the document gives us two hints. If there are races, which have the same race number they belong to different fleets. Or a regatta is structured into fleets if the race name has a second word in its name which represents the fleet color. We chose the second approach, so we do not have to make up the fleet colors and take them from the XRR document.

Process description

If you want to import a regatta structure of an event, you have to insert the URL of the JSON document from manage2sail.com. It is also possible to insert the event ID instead which is automatically transformed in the appropriate JSON URL then. The application parses the JSON document for the event as well as the XRR documents for each regatta. The UI displays all regattas of the event so that the user can select individually the regattas he wants to import. The user can also set default settings for a regatta structure. A regatta structure means for example all regattas, which are structured into Qualification, Finals and Medal. When editing the settings of a regatta structure, the user can edit settings of series structures. Series structure is the combination of fleets within a series. For example a Qualification structured into one fleet, or a Qualification structured into two fleets - yellow and blue.

If the user imports regattas into the system, the regattas receive the settings belonging to their regatta and series structure.

Details of the code

A new class `StructureImportManagementPanel` has been created in order to display the regatta structure import as a new tab in the Admin Console. This class passes the event url to the server, where the event is parsed. For the parsing and analyzing of regatta structures a new package `com.sap.sailing.xrr.structureImport` has been created. The main part of the analyzing does the class `StructureImporter` in this package. `StructureImporter` provides three public methods `parseEvent(String url)`, `getRegattas(Iterable<RegattaJSON> regattas)` and `setCompetitors(Set<RegattaResults> results, String boatClassName)`. The first method parses the event from the JSON URL. The result is an `Iterable` of `RegattaJSON`. `RegattaJSON` is a class, which holds the data of a regatta from the event JSON document. This includes the link to the XRR document and the regatta name for example. These regatta information are used to parse the regattas in the method `getRegattas(Iterable<RegattaJSON> regattas)`. It parses all regatta XRR documents in parallel to save time, because the download of the document is time expensive. After parsing the regattas, the method analyzes them in order to get their regatta and series structure. For this the class `BuildStructure` is used. It analyzes the series and fleets like written above. These heuristics are outsourced with a strategy pattern. The heuristic for series are implemented in `GuessSeriesFromRaceName`, which implements `GuessSeriesStrategy`. The strategy class is passed to the constructor of `RegattaStructure`. The heuristic for guessing the race number is also outsourced with a strategy pattern to `SetRacenumFromSeries`, which implements `SetRacenumStrategy`. This class is a parameter in the constructor of `StructureImporter`, because the race numbers are set in this class after the series have been analyzed. When the structure of a regatta has been analyzed, the method `getRegattas()` converts the regatta to an object of type `RegattaImpl`. For that default values are taken, which can be edited afterwards. After all regattas have been converted to `RegattaImpl` objects, the method returns those. These objects can be displayed in the UI in order to select them.

When `StructureImportManagementPanel` receives the regattas, it analyzes them in terms of their series structure. The fleet structure is analyzed in the dialog to edit the regattas with this series structure. By editing the default settings for this series structure the settings are written directly into the regatta objects displayed in the list in the UI.